

CURRICULUM VITAE

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Personal Information:

Date of Birth: 6th April, 1982
Citizenship: INDIAN
Gender: Male
Marital Status: Unmarried

Academic Background:

Ph. D.	Ph. D. in Theoretical Chemistry	University of Hyderabad Hyderabad, India	Supervisor: Prof Susanta Mahapatra
2005	M.Sc. in Physical Chemistry	Visva-Bharati, INDIA	
2003	B.Sc. in Chemistry (Hons)	Vidyasagar University, INDIA	

Doctoral Research:

Thesis Title: Quantum Mechanical Studies of Nonadiabatic Molecular Processes.

Brief description of my research work:

Photophysics and multi-state multi-mode vibronic dynamics on coupled electronic states (photo-ionized or photo-excited) of polyatomic molecular processes are investigated with the aid of *ab initio* electronic structure calculation and quantum dynamical methods. In order to examine the nonadiabatic effects in molecular processes, model diabatic Hamiltonians (in order to avoid the singular nonadiabatic coupling of the adiabatic electronic basis) are constructed. The Hamiltonians are written in terms of dimensionless normal displacement coordinates of vibrational modes. The coupling parameters of the electronic part of the diabatic Hamiltonian are extracted by performing rigorous *ab initio* calculations of electronic potential energy surfaces (PESs). I have used different quantum chemistry methods like Second order Moller-Plesset perturbation theory (MP2), Outer valence Green function method (OVGF), Complete active space self consistent field (CASSCF), Multi reference configuration interaction (MRCI), Equation of motion coupled cluster singles and doubles EOM-CCSD), etc for the electronic structure calculations e.g geometry optimization, frequency calculation, vertical ionization or excitation energy calculation. The conical intersections (CIs) of PESs are analyzed by both *ab initio* calculations and vibronic coupling (VC) theory. I have estimated the location of the minima of the seam of CIs relative to the minima of excited electronic states and the Jahn-Teller (JT) stabilization energies. The theoretically generated photo-ionized or photo-excited spectrum are probed through various modern spectroscopic techniques like photo-electron spectra, matrix isolation spectrum, vacuum ultraviolet absorption spectrum in the experiment. The ultrafast relaxation time of excited electronic states are theoretically estimated and compared with the experimental results.

The quantum dynamical observables are calculated by solving the time-independent as well as time-dependent Schrödinger equation. In the time-independent case, Lanczos algorithm is used to diagonalize the Hamiltonian matrix expressed in the basis of harmonic oscillator functions. This matrix diagonalization approach becomes computationally very expensive and often intractable for systems with large number of electronic and vibrational degrees of freedom (DOF). The wave packet (WP) propagation approach within the multi-configuration time-dependent Hartree (MCTDH) scheme has been employed for such situations. The rate of ultrafast nonradiative deactivation of excited electronic states is estimated from the time-dependence of diabatic and adiabatic electronic population during the evolution of WP.

The static and dynamic aspect of JT and PJT interactions in the excited electronic states of methyl cyanide and 2,2,2-bicyclo-octatriene radical cation are investigated. Static and dynamic aspects of electronically excited anthracene radical cation as archetypal model for astrophysical observations are examined. The photophysics of low-lying coupled excited electronic states of pyrimidine molecule and its radical cation is investigated.

The geometry and optical properties of series of 4,4'- π -conjugated-2,2'-bipyridine chromophores have been analyzed by DFT and TD-DFT computation in solvent model. The geometry optimization of isomeric calix[4]pyrroles have been performed by DFT method.

Project Experiences:

- Theoretical Investigation of Molecular Modeling on Promising High Energy Density Molecules (HEDM).

Computer Knowledge:

Language: Fortran 77
Operating Systems: Linux, Windows

Quantum Chemistry Packages known:

Gaussian 98/03, Molpro, MCTDH, Turbomole, Mopac, Material Studio, Molden, Molekel, Maple etc.

Awards and Fellowships:

- GATE Qualified in 2005.
- CSIR Junior Research Fellowship (JRF) – by qualifying the All India National Eligibility Test (NET) conducted by CSIR-UGC, INDIA, in 2006.

Publications:

11. **S. Ghanta** and S. Mahapatra, “Theoretical investigation of the photo-stability of pyrimidine molecule and its radical cation” (to be submitted).

10. **S. Ghanta** and S. Mahapatra, “Photophysics of barrelene: The Jahn-Teller and pseudo-Jahn-Teller effects” (to be submitted).

9. R. Samanta, S. P. Mahanta, **S. Ghanta** and P. K. Panda, “Naphthalene strapped fluorescent calix[4]pyrrole isomers: halide ion selectivity based on strap topography”. *RSC Advances*, **2**, 7974-7977 (2012).

8. M. Sarma, T. Chatterjee, **S. Ghanta** and S. K. Das, "D- π -A-A- π -D Prototype 2,2'-Bipyridine Dyads Exhibiting Large Structure and Environment Sensitive Fluorescence: Synthesis, Photophysics and Computation". *J. Org. Chem.* **77**, 432-444 (2012).

7. T. Chatterjee, M. Sarma, **S. Ghanta** and S. K. Das, “*Sterically driven electronic properties of naphthalene- and anthracene-end-capped 2,2'-bipyridine luminophores: Synthesis and density functional theory*”. *Tetrahedron. Lett.* **52**, 5460-5463 (2011).

6. **S. Ghanta**, V. Sivaranjana Reddy and S. Mahapatra, "*Theoretical study of electronically excited radical cations of naphthalene and anthracene as archetypal models for astrophysical observations. Part II. Dynamics Consequences.*" *Phys. Chem. Chem. Phys.*, **13**, 14531-14541 (2011).

5. **S. Ghanta**, V. Sivaranjana Reddy and S. Mahapatra, "*Theoretical study of electronically excited radical cations of naphthalene and anthracene as archetypal models for astrophysical observations. Part I. Static aspects.*" *Phys. Chem. Chem. Phys.* **13**, 14523 - 14530 (2011).

4. V. Sivaranjana Reddy, **S. Ghanta** and S. Mahapatra, "*First principles quantum dynamical investigation provides evidence for the role of polycyclic aromatic hydrocarbon radical cations in interstellar physics.*" *Phys. Rev. Lett.*, **104**, 111102(1-4), (2010).

3. T. Mondal, B. Saritha, **S. Ghanta**, T. K. Roy, S. Mahapatra, and M. Durga Prasad, "*On some strategies to design new high energy density molecules*", *J. Mol. Struct.(THEOCHEM.)* **897**, 42-47, (2008).

2. **S. Ghanta** and S. Mahapatra, " *Vibronic dynamics in the low-lying coupled electronic states of methyl cyanide radical cation*". *Chem. Phys.* **347**, 97-109 (2008).

1. Tapta Kanchan Roy, **Susanta Ghanta**, Tanmoy Mondal, Banda Saritha, S. Mahapatra, and M. Durga Prasad, "*Conformational preferences of mono substituted cyclohydronitrogens : A Theoretical study*". *J. Mol. Struct.(THEOCHEM.)* **822**, 145-150 (2007).

Conference and Presentation:

- Poster presented in "**Theoretical Chemistry Symposium (TCS 2006)**", December 11-13, 2006, Bharathidasan University, Thiruchirappalli. India.
- Poster presented in "**Chemfest 2007**", March 9-10, 2007, School of Chemistry, University of Hyderabad, Hyderabad, India.
- Poster presented in "**Spectroscopy and Dynamics of Molecules and Clusters (SDMC 2008)**", February 22-24, 2008 Mahabalipuram. Tamil Nadu, India.
- Poster presented in "**Chemfest 2008**", March 1-2, 2008, School of Chemistry, University of Hyderabad, Hyderabad, India.
- Poster presented in "**Spectroscopy and Dynamics of Molecules and Clusters (SDMC 2009)**", February 20-22, 2009 Mandarmoni, West Bengal, India.
- **Oral Presentation** in "**Chemfest 2009**", March 7-8, 2009, School of Chemistry, University of Hyderabad, Hyderabad, India.

- Poster presented in "**Chemfest 2010**", January 8-9, 2010, School of Chemistry, University of Hyderabad, Hyderabad, India.
- "Poster presented in *XXth International Symposium on the Jahn-Teller effect*" August 16-20, 2010, **University of Fribourg, Fribourg, Switzerland**.
- Poster presented in "**Spectroscopy and Dynamics of Molecules and Clusters (SDMC 2011)**", February 19-21, 2011, Indian Institute of Technology Kanpur, Kanpur, India.

Referees:-

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